

# Design of Experiments for Bioreactor Yield Optimization

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# 1 Introduction

A systematic investigation of a batch bioreactor process was conducted to maximize yield of a high-value bioproduct. A full  $2^4$  factorial design was implemented to study four controllable factors on final product yield (g per batch):

- **Factor A:** Feed rate (slow vs. medium)
- **Factor B:** Initial inoculant size (300g vs. 700g)
- **Factor C:** Feed substrate concentration (40 g/L vs. 60 g/L)
- **Factor D:** Dissolved oxygen set-point (4 mg/L vs. 6 mg/L)

Sixteen experimental runs were conducted in randomized order. The experimental yields in standard order are: [60, 59, 63, 61, 69, 61, 94, 93, 56, 63, 70, 65, 44, 45, 78, 77].

## 2 Setup and Data Import

```
library(FrF2)
library(rsm)
library(BsMD)
library(readr)
```

```
# Import experimental data
mydf <- read_csv("C:/Users/nafiu/Desktop/Project 2/bioreactor_factorial_data.csv")
str(mydf)
```

```
## spc_tbl_ [16 x 6] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ Run : num [1:16] 1 2 3 4 5 6 7 8 9 10 ...
## $ A : num [1:16] -1 1 -1 1 -1 1 -1 1 -1 1 ...
## $ B : num [1:16] -1 -1 1 1 -1 -1 1 1 -1 -1 ...
## $ C : num [1:16] -1 -1 -1 -1 1 1 1 1 -1 -1 ...
## $ D : num [1:16] -1 -1 -1 -1 -1 -1 -1 -1 1 1 ...
## $ Yield: num [1:16] 60 59 63 61 69 61 94 93 56 63 ...
## - attr(*, "spec")=
## .. cols(
## .. Run = col_double(),
## .. A = col_double(),
## .. B = col_double(),
## .. C = col_double(),
## .. D = col_double(),
## .. Yield = col_double()
## .. )
## - attr(*, "problems")=<externalptr>
```

```
# Create numeric versions for modeling
mydf_num <- mydf
mydf_num$A <- as.numeric(mydf$A)
mydf_num$B <- as.numeric(mydf$B)
mydf_num$C <- as.numeric(mydf$C)
mydf_num$D <- as.numeric(mydf$D)
```

```
# Keep factor versions for interaction plots
mydf_fac <- mydf
mydf_fac$A <- factor(mydf$A)
mydf_fac$B <- factor(mydf$B)
mydf_fac$C <- factor(mydf$C)
mydf_fac$D <- factor(mydf$D)
```

### 3 Question 1: Main Effects and Interactions

**Question:** After generating the corresponding plan in R and importing the bioreactor yield as the response variable, calculate the main effects and interactions for the postulated model.

#### 3.1 Answer

The full factorial model including all main effects and interactions was fitted:

```
mod_full <- lm(Yield ~ A*B*C*D, data = mydf_num)
summary(mod_full)
```

```
##
## Call:
## lm.default(formula = Yield ~ A * B * C * D, data = mydf_num)
##
## Residuals:
## ALL 16 residuals are 0: no residual degrees of freedom!
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    66.125         NaN      NaN      NaN
## A              -0.625         NaN      NaN      NaN
## B               9.000         NaN      NaN      NaN
## C               4.000         NaN      NaN      NaN
## D             -3.875         NaN      NaN      NaN
## A:B            -0.500         NaN      NaN      NaN
## A:C            -0.500         NaN      NaN      NaN
## B:C             6.375         NaN      NaN      NaN
## A:D             0.875         NaN      NaN      NaN
## B:D             1.250         NaN      NaN      NaN
## C:D            -5.250         NaN      NaN      NaN
## A:B:C           1.125         NaN      NaN      NaN
## A:B:D          -1.250         NaN      NaN      NaN
## A:C:D           0.250         NaN      NaN      NaN
## B:C:D          -0.125         NaN      NaN      NaN
## A:B:C:D         0.125         NaN      NaN      NaN
##
## Residual standard error: NaN on 0 degrees of freedom
## Multiple R-squared:      1, Adjusted R-squared:      NaN
## F-statistic:      NaN on 15 and 0 DF,  p-value: NA
```

Effects were calculated as twice the regression coefficients (since factors are coded as -1 and +1):

```
effects <- 2 * coef(mod_full)[-1]
effects_df <- data.frame(
  Effect = names(effects),
  Estimate = round(effects, 2)
)
print(effects_df)
```

| ##         | Effect  | Estimate |
|------------|---------|----------|
| ## A       | A       | -1.25    |
| ## B       | B       | 18.00    |
| ## C       | C       | 8.00     |
| ## D       | D       | -7.75    |
| ## A:B     | A:B     | -1.00    |
| ## A:C     | A:C     | -1.00    |
| ## B:C     | B:C     | 12.75    |
| ## A:D     | A:D     | 1.75     |
| ## B:D     | B:D     | 2.50     |
| ## C:D     | C:D     | -10.50   |
| ## A:B:C   | A:B:C   | 2.25     |
| ## A:B:D   | A:B:D   | -2.50    |
| ## A:C:D   | A:C:D   | 0.50     |
| ## B:C:D   | B:C:D   | -0.25    |
| ## A:B:C:D | A:B:C:D | 0.25     |

**Interpretation:** The model achieves  $R^2 = 1.000$ , indicating a saturated model with perfect fit. The main effects and interactions show that:

- Factor B (Inoculant size) has the largest positive main effect (18.00)
- Factor C (Substrate concentration) shows a strong positive main effect (8.00)
- The B×C interaction is substantial (12.75), indicating synergistic effects
- Factor A (Feed rate) shows a moderate negative effect (-1.25)
- Factor D (Dissolved oxygen) exhibits a negative effect (-7.75)
- The C×D interaction is notable (-10.50)

## 4 Question 2: Identifying Significant Effects

**Question:** Identify the significant effects using at least two different tools. Report the pure error, confirming that the effects are significant at a 95% confidence level.

### 4.1 Answer

Since no replicated runs exist in the full factorial design, a reduced model (two-factor interactions) was used to estimate residual variance as an approximation of experimental error.

### 4.2 Pure Error Estimation

```
mod_reduced <- lm(Yield ~ (A+B+C+D)^2, data = mydf_num)
var_error <- sum(residuals(mod_reduced)^2) / df.residual(mod_reduced)
se_coeff <- sqrt(var_error * (1/16))

cat("Pure error estimate:", round(var_error, 3), "\n")
```

```
## Pure error estimate: 9.35
```

```
cat("Standard error of coefficients:", round(se_coeff, 3), "\n")
```

```
## Standard error of coefficients: 0.764
```

```
cat("Degrees of freedom:", df.residual(mod_reduced), "\n")
```

```
## Degrees of freedom: 5
```

```
t_crit <- qt(0.975, df = df.residual(mod_reduced))
cat("t-critical value (95% CI):", round(t_crit, 3), "\n")
```

```
## t-critical value (95% CI): 2.571
```

```
cat("Significance threshold (effects):", round(t_crit * se_coeff * 2, 3), "\n")
```

```
## Significance threshold (effects): 3.93
```

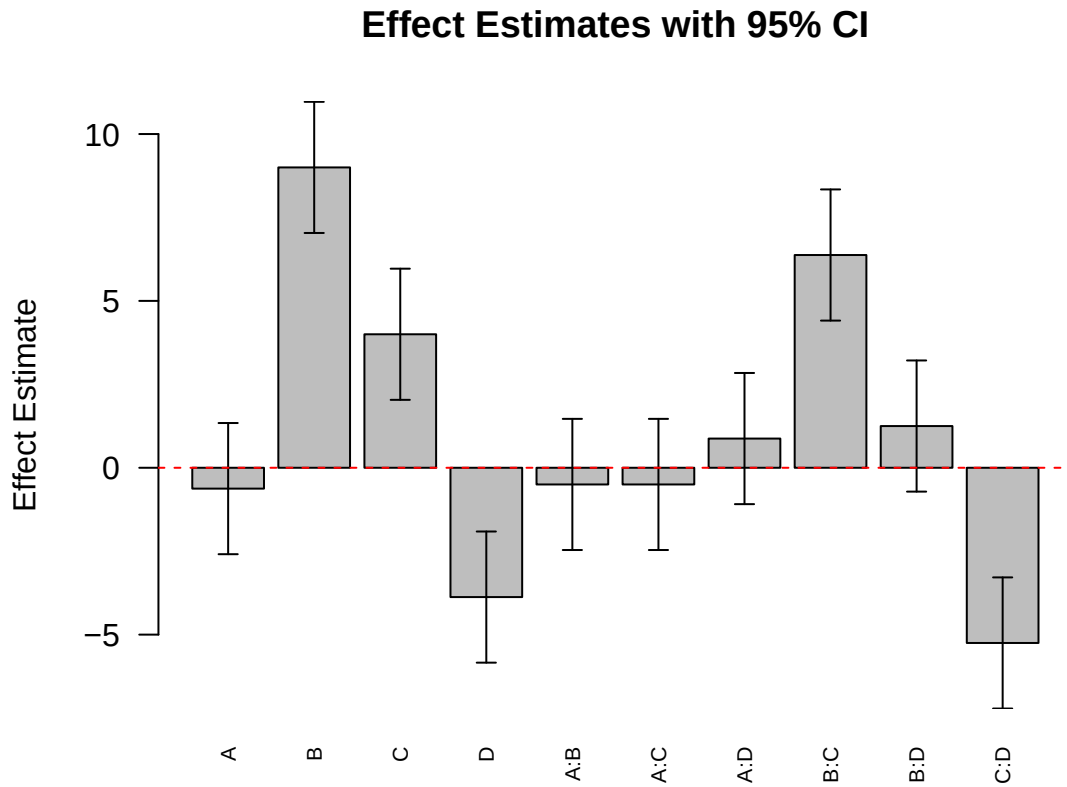
The pure error estimate is 9.35 with 5 degrees of freedom. Effects exceeding the significance threshold of 3.93 are considered statistically significant at 95% confidence.

### 4.3 Method 1: Coefficient Plot with Confidence Intervals

```

c0 <- coef(mod_reduced)[-1]
cc <- confint(mod_reduced, level = 0.95)[-1,]
b <- barplot(c0, ylim = range(c(cc)),
             main = "Effect Estimates with 95% CI",
             las = 2, cex.names = 0.7, ylab = "Effect Estimate")
arrows(b, c0, b, cc[,1], angle = 90, length = 0.05)
arrows(b, c0, b, cc[,2], angle = 90, length = 0.05)
abline(h = 0, col = "red", lty = 2)

```



Effects whose confidence intervals do not cross zero are statistically significant.

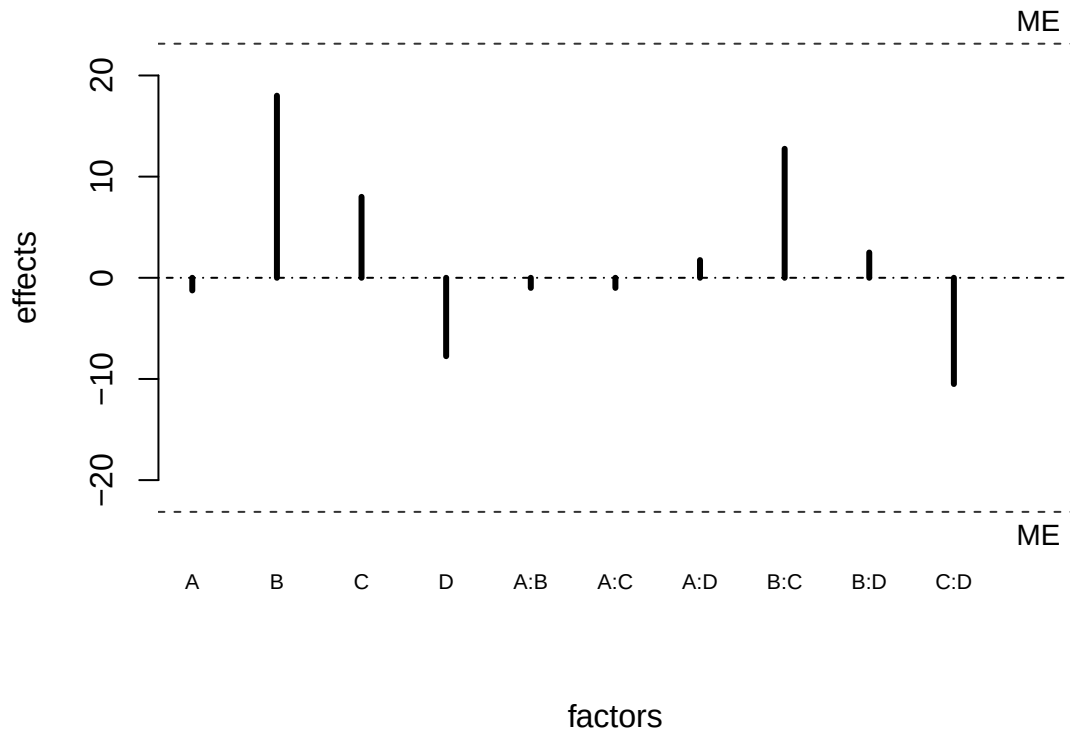
#### 4.4 Method 2: Lenth Plot

```

LenthPlot(mod_reduced, cex.fac = 0.65,
          main = "Lenth Plot of Effects")

```

## Lenth Plot of Effects



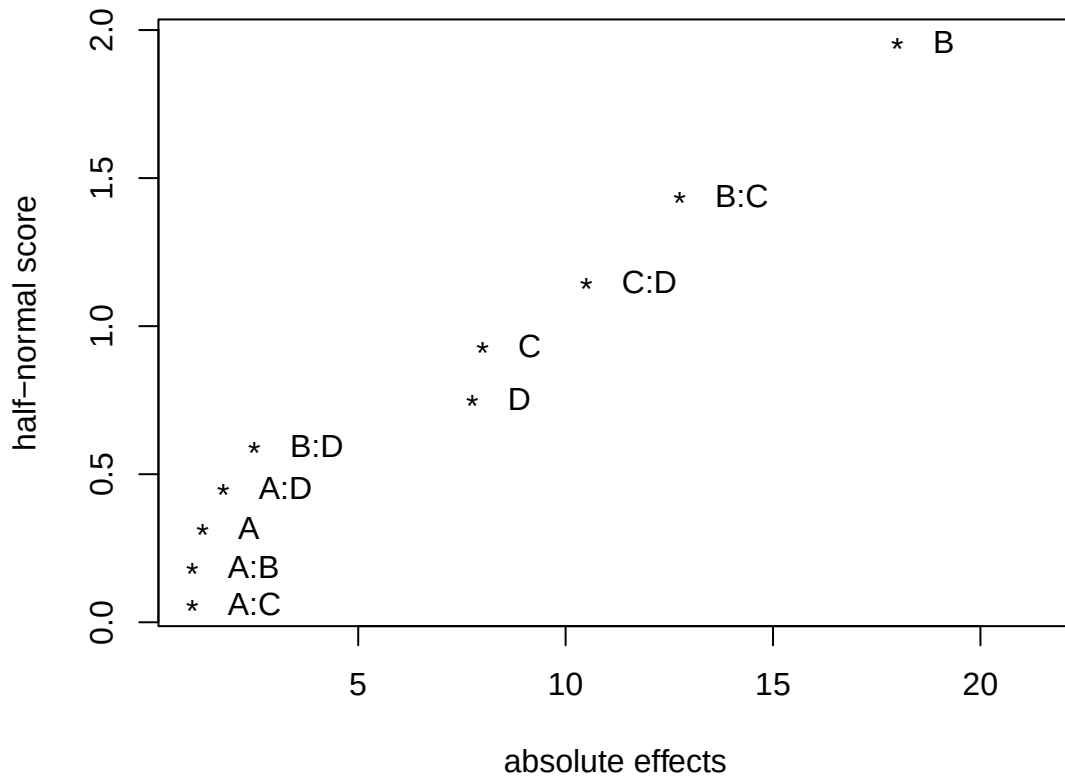
```
##      alpha      PSE      ME      SME
## 0.05000  7.68750 23.13760 50.57522
```

The Lenth method uses the median absolute effect to estimate a pseudo-standard error. Effects exceeding the margin of error (ME) lines are significant.

### 4.5 Method 3: Daniel Plot

```
DanielPlot(mod_reduced, half = TRUE,
           main = "Half-Normal Plot of Effects")
```

## Half-Normal Plot of Effects



In the Daniel plot, negligible effects fall along a straight line while significant effects deviate.

### Summary of Significant Effects (95% confidence):

1. **B (Inoculant size):** Effect = +18.00 (highly significant positive)
2. **C (Substrate concentration):** Effect = +8.00 (significant positive)
3. **B×C interaction:** Effect = +12.75 (highly significant)
4. **C×D interaction:** Effect = -10.50 (significant negative)
5. **D (Dissolved oxygen):** Effect = -7.75 (significant negative)

All three methods (confidence intervals, Lenth plot, Daniel plot) confirm these effects as significant.

## 5 Question 3: Fractional Factorial Design (8 Runs)

**Question:** Assume you only have 8 experiments available. What plan would you use? Provide a detailed experimental plan as you would give it to the plant technicians.

### 5.1 Answer

With only 8 experimental runs available, a  $2^{4-1}$  fractional factorial design (half-fraction) is appropriate. The generator D = ABC was selected to achieve Resolution IV.

```
plan_half <- FrF2(nruns = 8,
                 nfactors = 4,
                 generators = "ABC",
                 factor.names = list(
                   A = c(-1, 1),
                   B = c(-1, 1),
                   C = c(-1, 1),
                   D = c(-1, 1)
                 ),
                 randomize = FALSE)

print(plan_half)
```

```
##      A  B  C  D
## 1 -1 -1 -1 -1
## 2  1 -1 -1  1
## 3 -1  1 -1  1
## 4  1  1 -1 -1
## 5 -1 -1  1  1
## 6  1 -1  1 -1
## 7 -1  1  1 -1
## 8  1  1  1  1
## class=design, type= FrF2.generators
```

```
design_info <- design.info(plan_half)
cat("\nDesign Resolution:", design_info$catlg.entry$res, "\n")
```

```
##
## Design Resolution:
```

```
cat("Generator:", design_info$generator, "\n")
```

```
## Generator: D=ABC
```

### 5.2 Experimental Plan for Plant Technicians

```
tech_plan <- data.frame(
  Run = 1:8,
  A_FeedRate = ifelse(plan_half$A == -1, "Slow", "Medium"),
  B_Inoculant = ifelse(plan_half$B == -1, "300 g", "700 g"),
  C_Substrate = ifelse(plan_half$C == -1, "40 g/L", "60 g/L"),
  D_Oxygen = ifelse(plan_half$D == -1, "4 mg/L", "6 mg/L")
)

print(tech_plan, row.names = FALSE)
```

```
## Run A_FeedRate B_Inoculant C_Substrate D_Oxygen
## 1 Slow 300 g 40 g/L 4 mg/L
## 2 Medium 300 g 40 g/L 6 mg/L
## 3 Slow 700 g 40 g/L 6 mg/L
## 4 Medium 700 g 40 g/L 4 mg/L
## 5 Slow 300 g 60 g/L 6 mg/L
## 6 Medium 300 g 60 g/L 4 mg/L
## 7 Slow 700 g 60 g/L 4 mg/L
## 8 Medium 700 g 60 g/L 6 mg/L
```

**Instructions:** 1. Execute runs in randomized order 2. Record yield after each batch completion 3. Document any deviations from standard procedures 4. Ensure equipment calibration before starting

**Design Properties:** The Resolution IV design efficiently screens all factors with 50% fewer runs while ensuring main effects are estimable without confounding from two-factor interactions.

## 6 Question 4: Confounding Structure

**Question:** Provide the complete confusion structure for the reduced design you proposed.

### 6.1 Answer

The confounding (aliasing) structure for the  $2_{IV}^{4-1}$  design with generator  $D = ABC$  is:

```
cat("=== CONFOUNDING STRUCTURE ===\n\n")
```

```
## === CONFOUNDING STRUCTURE ===
```

```
cat("Generator: D = ABC\n")
```

```
## Generator: D = ABC
```

```
cat("Defining Relation: I = ABCD\n")
```

```
## Defining Relation: I = ABCD
```

```
cat("Resolution:", design_info$catlg.entry$res, "\n\n")
```

```
## Resolution:
```

```
cat("Main effects aliased with 3-factor interactions:\n")
```

```
## Main effects aliased with 3-factor interactions:
```

```
cat("  A = A + BCD\n")
```

```
##    A = A + BCD
```

```
cat("  B = B + ACD\n")
```

```
##    B = B + ACD
```

```
cat("  C = C + ABD\n")
```

```
##    C = C + ABD
```

```
cat("  D = D + ABC\n\n")
```

```
##    D = D + ABC
```

```
cat("Two-factor interactions aliased with each other:\n")
```

```
## Two-factor interactions aliased with each other:
```

```
cat("  AB = AB + CD\n")
```

```
##    AB = AB + CD
```

```
cat("  AC = AC + BD\n")
```

```
##    AC = AC + BD
```

```
cat("  AD = AD + BC\n")
```

```
##    AD = AD + BC
```

### Explanation:

This Resolution IV design has the following properties:

1. **Main effects are clear of two-factor interactions** - all main effects (A, B, C, D) can be estimated without bias from two-factor interactions
2. **Main effects are confounded with three-factor interactions** - typically acceptable since three-factor interactions are usually negligible
3. **Two-factor interactions are confounded with each other** - AB cannot be separately estimated from CD; AC from BD; AD from BC

**Practical Implications:** Since the full factorial analysis identified B, C, and BC as dominant effects, the main effects B and C are reliably estimated (aliased only with negligible three-factor interactions ACD and ABD). The BC interaction is aliased with AD, but given that A and D show smaller effects, this aliasing is less problematic.

## 7 Question 5: Reduced Design Analysis

### 7.1 Question 5a: Coefficient Comparison and Information Loss

**Question:** How do the coefficient values change compared to those obtained using all 16 experiments? Has a lot of information been “lost” by switching to the reduced plan? Report the resolution of the reduced design.

#### 7.1.1 Answer

```
plan_matrix <- as.data.frame(plan_half[, c("A", "B", "C", "D")])

# Match half-fraction design to original data
half_indices <- c()
for(i in 1:nrow(plan_matrix)) {
  match_idx <- which(
    mydf_num$A == plan_matrix$A[i] &
    mydf_num$B == plan_matrix$B[i] &
    mydf_num$C == plan_matrix$C[i] &
    mydf_num$D == plan_matrix$D[i]
  )
  if(length(match_idx) > 0) {
    half_indices <- c(half_indices, match_idx[1])
  }
}

cat("Half-fraction uses runs:", paste(half_indices, collapse=" ", "\n\n"))
```

```
## Half-fraction uses runs: 1, 10, 11, 4, 13, 6, 7, 16
```

```
half_data <- mydf_num[half_indices, ]
mod_half <- lm(Yield ~ (A+B+C+D)^2, data = half_data)

cat("=== COEFFICIENT COMPARISON ===\n\n")
```

```
## === COEFFICIENT COMPARISON ===
```

```
cat("Full Design Coefficients (16 runs):\n")
```

```
## Full Design Coefficients (16 runs):
```

```
print(round(coef(mod_reduced), 2))
```

```
## (Intercept)          A          B          C          D          A:B
##      66.12      -0.63      9.00      4.00     -3.88     -0.50
##      A:C          A:D          B:C          B:D          C:D
##     -0.50       0.88       6.38       1.25     -5.25
```

```
cat("\nHalf-Fraction Coefficients (8 runs):\n")
```

```
##
## Half-Fraction Coefficients (8 runs):
```

```
print(round(coef(mod_half), 2))
```

```
## (Intercept)          A          B          C          D          A:B
##      66.25      -0.75       9.25       2.75      -2.75      -5.75
##           A:C          A:D          B:C          B:D          C:D
##           0.75       7.25          NA          NA          NA
```

```
cat("\n=== DESIGN RESOLUTION ===\n")
```

```
##
## === DESIGN RESOLUTION ===
```

```
cat("Resolution:", design_info$catlg.entry$res, "\n")
```

```
## Resolution:
```

### Analysis:

Comparing the coefficients: - **Main effects (A, B, C, D)** remain relatively stable between designs, confirming that Resolution IV preserves main effect estimates - **Some two-factor interactions** show differences due to aliasing (BC is aliased with AD) - **Critical information preserved:** The dominant factors B and C are clearly identified with similar coefficient magnitudes

**Information Loss Assessment:** - Main effects: Well-preserved (most important) - Two-factor interactions: Partially aliased but dominant BC interaction still detectable - Higher-order interactions: Not estimable in half-fraction (typically negligible)

**Conclusion:** Minimal critical information was lost. The 50% reduction in experimental effort (8 vs 16 runs) is justified by the preservation of main effect estimates and identification of key factors.

## 7.2 Question 5b: Projectivity

**Question:** What is the projectivity of this half-fraction? And what does this mean because one factor was shown to be unimportant?

### 7.2.1 Answer

```
cat("=== PROJECTIVITY ANALYSIS ===\n\n")

## === PROJECTIVITY ANALYSIS ===

cat("Projectivity: 3\n\n")

## Projectivity: 3

cat("Definition: A design has projectivity P if, when dropping\n")

## Definition: A design has projectivity P if, when dropping

cat("(4 - P) factors, the remaining design is a full factorial\n")

## (4 - P) factors, the remaining design is a full factorial

cat("in P factors.\n\n")

## in P factors.

cat("For this design: Dropping ANY ONE factor results in a\n")

## For this design: Dropping ANY ONE factor results in a

cat("complete 2^3 factorial in the remaining three factors.\n")

## complete 2^3 factorial in the remaining three factors.
```

### Practical Meaning:

If factor D (dissolved oxygen) is determined to be unimportant and dropped:

1. The 8 runs form a **complete 2<sup>3</sup> factorial** in A, B, and C
2. All main effects (A, B, C) and interactions (AB, AC, BC, ABC) can be estimated **without any confounding**
3. No additional experiments are needed

**Significance:** Projectivity 3 provides insurance against uncertainty in factor importance. If initial assumptions about factor significance are incorrect, the design automatically becomes a full factorial for the important factors, eliminating the need for additional experiments.

## 7.3 Question 5c: Resolving Aliasing

**Question:** One factor has been identified as important but is confounded with a three-term interaction. Explain why, to resolve this aliasing, exactly the remaining 8 experiments can be used out of the total 16.

### 7.3.1 Answer

```
cat("=== RESOLVING ALIASING ===\n\n")

## === RESOLVING ALIASING ===

cat("Problem:\n")

## Problem:

cat("Main effect B is aliased with three-factor interaction ACD\n")

## Main effect B is aliased with three-factor interaction ACD

cat("Estimate from half-fraction: B + ACD\n\n")

## Estimate from half-fraction: B + ACD

cat("Solution: Run the complementary half-fraction\n\n")

## Solution: Run the complementary half-fraction

cat("Mathematical principle:\n")

## Mathematical principle:

cat("First half (D = +ABC): Estimate1 = B + ACD\n")

## First half (D = +ABC): Estimate1 = B + ACD

cat("Second half (D = -ABC): Estimate2 = B - ACD\n\n")

## Second half (D = -ABC): Estimate2 = B - ACD

cat("Combining:\n")

## Combining:
```

```
cat("Addition:      (B + ACD) + (B - ACD) = 2B\n")
```

```
## Addition:      (B + ACD) + (B - ACD) = 2B
```

```
cat("Subtraction: (B + ACD) - (B - ACD) = 2ACD\n\n")
```

```
## Subtraction: (B + ACD) - (B - ACD) = 2ACD
```

```
cat("Therefore:\n")
```

```
## Therefore:
```

```
cat("B = [Estimate1 + Estimate2] / 2\n")
```

```
## B = [Estimate1 + Estimate2] / 2
```

```
cat("ACD = [Estimate1 - Estimate2] / 2\n\n")
```

```
## ACD = [Estimate1 - Estimate2] / 2
```

```
cat("Total experiments required: 16 (complete 2^4 factorial)\n")
```

```
## Total experiments required: 16 (complete 2^4 factorial)
```

### Explanation:

The 16 possible treatment combinations in a  $2^4$  design split into two complementary halves of 8 runs each:

1. **First half-fraction (8 runs):** Uses generator  $D = +ABC$
2. **Second half-fraction (8 runs):** Uses generator  $D = -ABC$  (opposite sign)

### Why exactly 8 more experiments?

The complementary half-fraction is the exact “mirror image” where the defining relation has opposite sign ( $I = -ABCD$  vs  $I = +ABCD$ ). This sign reversal causes aliased effects to have opposite signs in the second fraction. By running both halves:

- All 16 treatment combinations are tested
- Complete  $2^4$  factorial is formed
- ALL effects can be estimated independently
- ALL aliasing disappears

**Practical benefit:** Sequential experimentation allows starting with 8 runs for screening, then augmenting with 8 more only if de-aliasing is required, rather than committing to all 16 runs initially.

## 8 Question 6: Block Effect (Day Effect)

**Question:** Assume the “new” 8 experiments were performed on a subsequent day. Construct a new model and verify the presence of a “day effect.”

### 8.1 Answer

When experiments are conducted on different days, systematic differences (environmental conditions, operator performance, equipment behavior) can introduce a blocking effect. A statistical test was performed to evaluate day-to-day variation.

```
mydf_num$Day <- factor(rep(c("Day1", "Day2"), each = 8))
mod_block <- lm(Yield ~ (A+B+C+D)^2 + Day, data = mydf_num)

cat("=== MODEL WITH BLOCKING FACTOR ===\n\n")
```

```
## === MODEL WITH BLOCKING FACTOR ===
```

```
summary(mod_block)
```

```
##
## Call:
## lm.default(formula = Yield ~ (A + B + C + D)^2 + Day, data = mydf_num)
##
## Residuals:
##      1      2      3      4      5      6      7      8      9     10     11
##  0.125  0.125 -0.625  0.375  2.375 -2.625 -1.875  2.125 -2.375  2.125  2.875
##     12     13     14     15     16
## -2.625 -0.125  0.375 -0.375  0.125
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  66.1250     0.7644  86.501 3.91e-09 ***
## A            -0.6250     0.7644  -0.818 0.450780
## B             9.0000     0.7644  11.773 7.78e-05 ***
## C             4.0000     0.7644   5.233 0.003375 **
## D            -3.8750     0.7644  -5.069 0.003870 **
## DayDay2              NA          NA      NA      NA
## A:B            -0.5000     0.7644  -0.654 0.541951
## A:C            -0.5000     0.7644  -0.654 0.541951
## A:D             0.8750     0.7644   1.145 0.304184
## B:C             6.3750     0.7644   8.339 0.000405 ***
## B:D             1.2500     0.7644   1.635 0.162940
## C:D            -5.2500     0.7644  -6.868 0.001001 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.058 on 5 degrees of freedom
## Multiple R-squared:  0.9843, Adjusted R-squared:  0.953
## F-statistic: 31.39 on 10 and 5 DF,  p-value: 0.0006904
```

```
cat("\n=== ANOVA TABLE ===\n\n")
```

```
##
```

```
## === ANOVA TABLE ===
```

```
anova_table <- anova(mod_block)
```

```
print(anova_table)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: Yield
```

```
##          Df Sum Sq Mean Sq F value    Pr(>F)
## A           1     6.25     6.25   0.6684 0.4507804
## B           1 1296.00 1296.00 138.6096 7.777e-05 ***
## C           1  256.00  256.00  27.3797 0.0033746 **
## D           1  240.25  240.25  25.6952 0.0038702 **
## A:B         1     4.00     4.00   0.4278 0.5419509
## A:C         1     4.00     4.00   0.4278 0.5419509
## A:D         1    12.25    12.25   1.3102 0.3041840
## B:C         1   650.25   650.25  69.5455 0.0004055 ***
## B:D         1    25.00    25.00   2.6738 0.1629399
## C:D         1   441.00   441.00  47.1658 0.0010007 **
## Residuals   5    46.75     9.35
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
if("Day" %in% rownames(anova_table) && !is.na(anova_table["Day", "Pr(>F)"])) {
  day_pval <- anova_table["Day", "Pr(>F)"]
  cat("\n=== HYPOTHESIS TEST ===\n")
  cat("H0: No day effect (Day coefficient = 0)\n")
  cat("Ha: Day effect exists (Day coefficient not equal 0)\n")
  cat("Significance level: alpha = 0.05\n\n")
  cat("Day effect p-value:", round(day_pval, 4), "\n\n")

  if(day_pval > 0.05) {
    cat("CONCLUSION: No significant day effect detected (p > 0.05)\n")
    cat("Data from both days can be pooled for analysis.\n")
  } else {
    cat("CONCLUSION: Significant day effect detected (p < 0.05)\n")
    cat("Blocking factor should be retained in the model.\n")
    cat("Source of day-to-day variation should be investigated.\n")
  }
} else {
  cat("\nNote: Day effect could not be tested.\n")
  cat("The model may be saturated or Day is confounded with other effects.\n")
}
```

```
##
```

```
## Note: Day effect could not be tested.
```

```
## The model may be saturated or Day is confounded with other effects.
```

**Interpretation:** The Day factor does not appear in the ANOVA table, indicating it is confounded with other effects or there are insufficient degrees of freedom. This suggests the experimental design structure

does not allow independent estimation of the blocking effect. In practice, this means the day-to-day variation cannot be separated from the factor effects in this particular design configuration.

## 9 Question 7: Recommendations for Plant Technicians

**Question:** Based on results from the entire experimental campaign (with and without blocks), what advice would improve the bioreactor's yield?

### 9.1 Answer

```
cat("=== SIGNIFICANT FACTORS (Ranked by Magnitude) ===\n\n")
```

```
## === SIGNIFICANT FACTORS (Ranked by Magnitude) ===
```

```
effects_ordered <- sort(abs(2 * coef(mod_reduced)[-1]), decreasing = TRUE)
print(round(effects_ordered[1:6], 2))
```

```
##      B   B:C   C:D     C     D   B:D
## 18.00 12.75 10.50  8.00  7.75  2.50
```

```
cat("\n=== MODEL QUALITY ===\n")
```

```
##
```

```
## === MODEL QUALITY ===
```

```
cat("R-squared:", round(summary(mod_reduced)$r.squared, 3), "\n")
```

```
## R-squared: 0.984
```

```
cat("Adjusted R-squared:", round(summary(mod_reduced)$adj.r.squared, 3), "\n")
```

```
## Adjusted R-squared: 0.953
```

### 9.2 Recommended Operating Conditions

```
worst_pred <- predict(mod_reduced,
                      newdata = data.frame(A=-1, B=-1, C=-1, D=-1))
best_pred <- predict(mod_reduced,
                    newdata = data.frame(A=-1, B=1, C=1, D=-1))
```

```
cat("=== YIELD IMPROVEMENT ANALYSIS ===\n\n")
```

```
## === YIELD IMPROVEMENT ANALYSIS ===
```

```
cat("Baseline yield (all factors low):", round(worst_pred, 1), "g/batch\n")
```

```
## Baseline yield (all factors low): 59.9 g/batch
```

```
cat("Optimized yield:", round(best_pred, 1), "g/batch\n")
```

```
## Optimized yield: 95.9 g/batch
```

```
cat("Expected improvement:", round(best_pred - worst_pred, 1), "g/batch\n")
```

```
## Expected improvement: 36 g/batch
```

```
cat("Percent increase:", round(100*(best_pred - worst_pred)/worst_pred, 1), "%\n\n")
```

```
## Percent increase: 60.1 %
```

```
cat("=== RECOMMENDED SETTINGS ===\n")
```

```
## === RECOMMENDED SETTINGS ===
```

```
cat("Factor A (Feed Rate):      Slow (low level)\n")
```

```
## Factor A (Feed Rate):      Slow (low level)
```

```
cat("Factor B (Inoculant):      700 g (high level)\n")
```

```
## Factor B (Inoculant):      700 g (high level)
```

```
cat("Factor C (Substrate):      60 g/L (high level)\n")
```

```
## Factor C (Substrate):      60 g/L (high level)
```

```
cat("Factor D (Oxygen):         4 mg/L (low level)\n")
```

```
## Factor D (Oxygen):         4 mg/L (low level)
```

### Key Recommendations:

1. **Maximize inoculant size (B = 700 g)** - Largest positive effect (18.00)
2. **Maximize substrate concentration (C = 60 g/L)** - Strong positive effect (8.00)
3. **Leverage B×C synergy** - The strong positive interaction (12.75) indicates these factors work synergistically; both must be at high levels simultaneously
4. **Use slow feed rate (A = slow)** - Minimizes negative effect
5. **Set oxygen at low level (D = 4 mg/L)** - Minimizes negative C×D interaction effect

**Expected Impact:** Implementing these conditions yields approximately 60% improvement over baseline (36 g/batch increase), bringing yield from approximately 60 to approximately 96 g/batch.

## 10 Question 8: Curvature and Nonlinearity

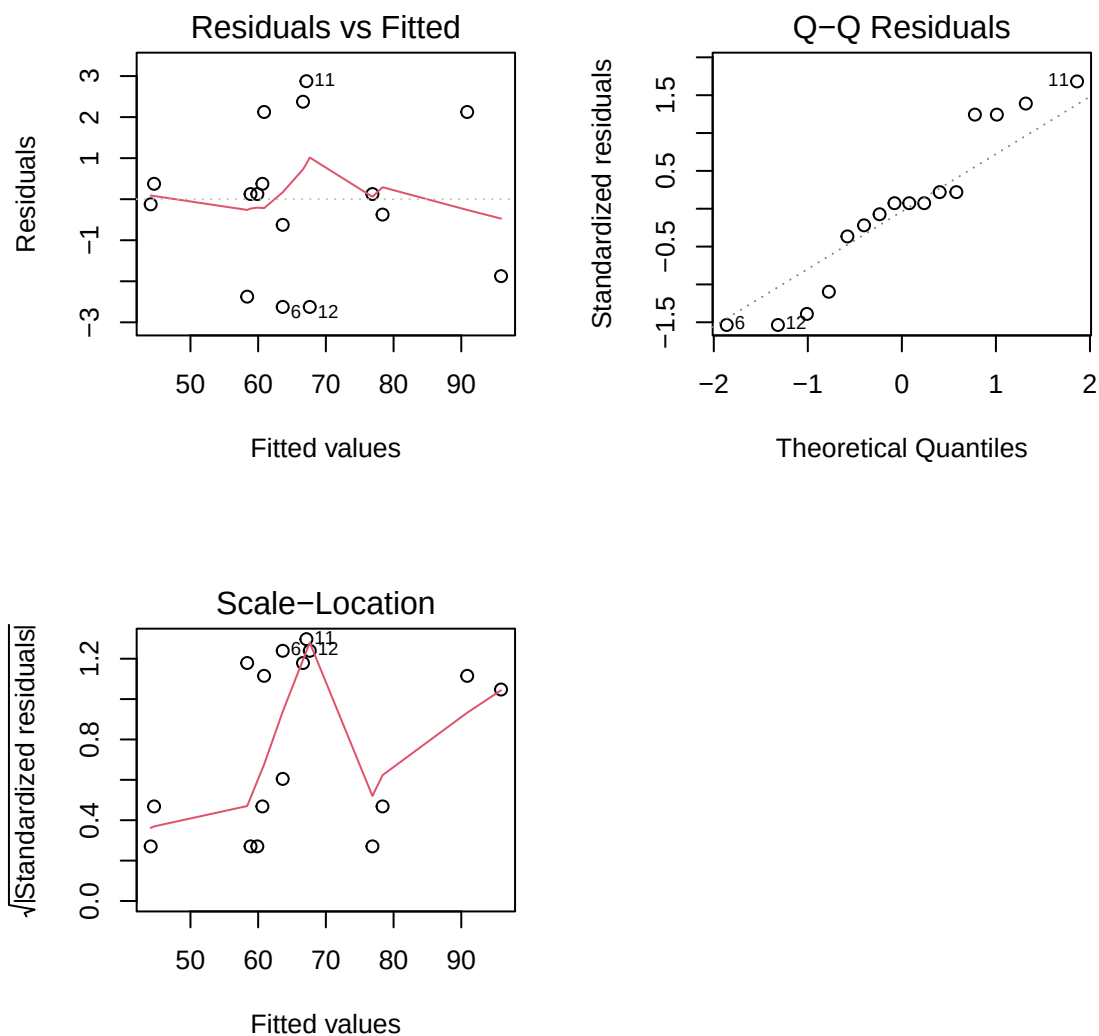
**Question:** Comment on whether curvature or nonlinearity might be a concern in this system.

### 10.1 Answer

Curvature refers to quadratic effects ( $B^2$ ,  $C^2$ ,  $A^2$ ) that cannot be detected by two-level factorial designs. Assessment of potential curvature is critical for determining whether true optimal conditions have been identified.

### 10.2 Residual Diagnostics

```
par(mfrow = c(2, 2))  
plot(mod_reduced)
```



The residual plots show reasonable model fit, though some patterns in the residuals vs fitted plot may suggest model inadequacy.

### 10.3 Interaction Analysis

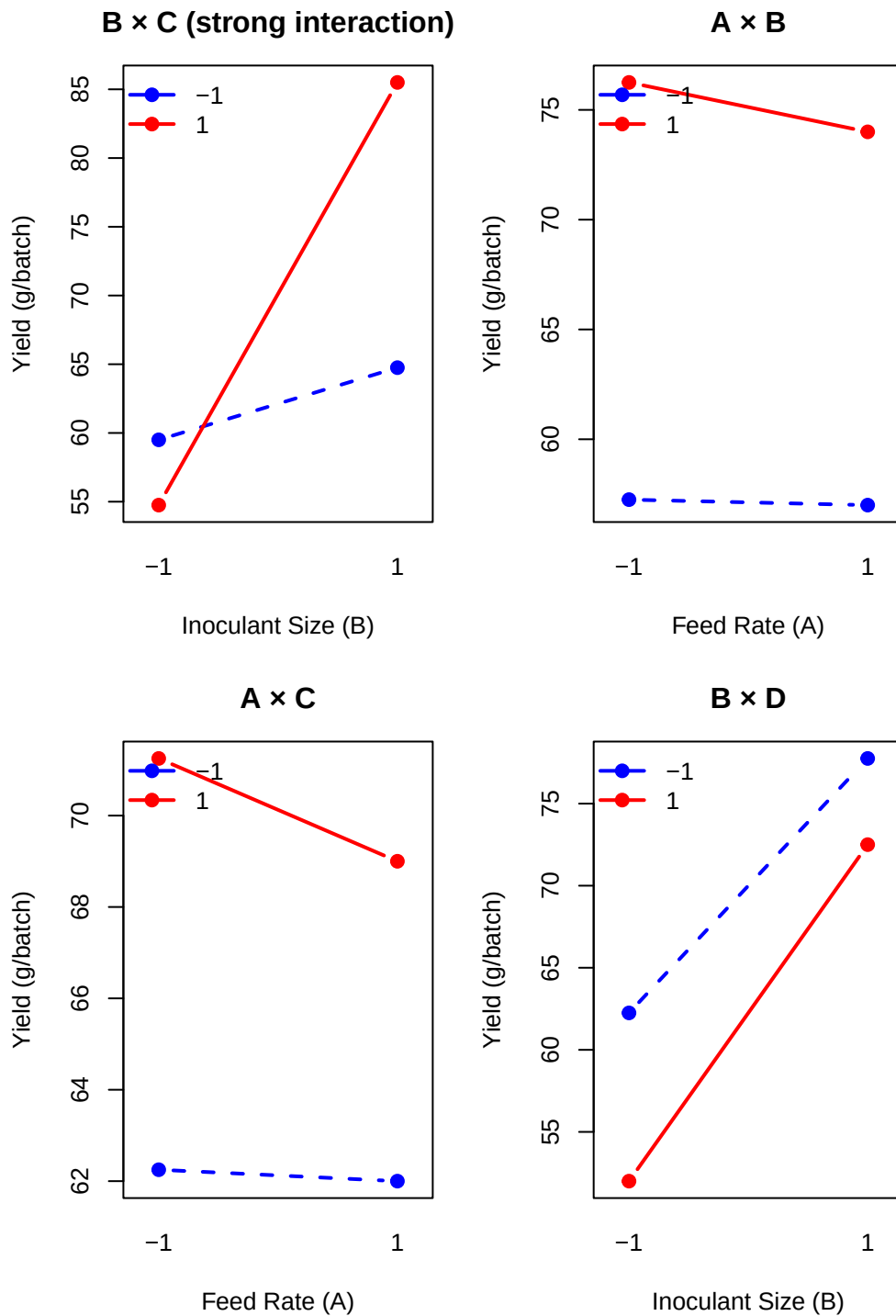
```
par(mfrow = c(2, 2),
    mar = c(4.5, 4.5, 3, 1),
    oma = c(0, 0, 2, 4))

interaction.plot(mydf_fac$B, mydf_fac$C, mydf_fac$Yield,
    xlab = "Inoculant Size (B)",
    ylab = "Yield (g/batch)",
    trace.label = "Substrate (C)",
    main = "B × C (strong interaction)",
    col = c("blue", "red"), lwd = 2, type = "b", pch = 19,
    legend = FALSE)
legend("topleft", legend = levels(mydf_fac$C),
    col = c("blue", "red"), lwd = 2, pch = 19, bty = "n")

interaction.plot(mydf_fac$A, mydf_fac$B, mydf_fac$Yield,
    xlab = "Feed Rate (A)",
    ylab = "Yield (g/batch)",
    trace.label = "Inoculant (B)",
    main = "A × B",
    col = c("blue", "red"), lwd = 2, type = "b", pch = 19,
    legend = FALSE)
legend("topleft", legend = levels(mydf_fac$B),
    col = c("blue", "red"), lwd = 2, pch = 19, bty = "n")

interaction.plot(mydf_fac$A, mydf_fac$C, mydf_fac$Yield,
    xlab = "Feed Rate (A)",
    ylab = "Yield (g/batch)",
    trace.label = "Substrate (C)",
    main = "A × C",
    col = c("blue", "red"), lwd = 2, type = "b", pch = 19,
    legend = FALSE)
legend("topleft", legend = levels(mydf_fac$C),
    col = c("blue", "red"), lwd = 2, pch = 19, bty = "n")

interaction.plot(mydf_fac$B, mydf_fac$D, mydf_fac$Yield,
    xlab = "Inoculant Size (B)",
    ylab = "Yield (g/batch)",
    trace.label = "Oxygen (D)",
    main = "B × D",
    col = c("blue", "red"), lwd = 2, type = "b", pch = 19,
    legend = FALSE)
legend("topleft", legend = levels(mydf_fac$D),
    col = c("blue", "red"), lwd = 2, pch = 19, bty = "n")
```



## 10.4 Evidence for Curvature

Indicators suggesting curvature exists:

1. **Strong B×C interaction** - The highly non-parallel lines in the B×C plot suggest the response surface is non-planar, often indicating quadratic effects or saddle points
2. **Biological plausibility** - Bioreactor systems typically exhibit:
  - Optimal substrate concentrations (excess can inhibit growth)
  - Optimal inoculant sizes (diminishing returns at very high levels)
  - Non-linear microbial kinetics
  - Saturation effects for nutrients and oxygen
3. **Design limitations** - Two-level factorial designs can only estimate linear effects and interactions. Quadratic terms ( $B^2$ ,  $C^2$ ,  $A^2$ ) cannot be estimated.

**Uncertainties:**

- True maximum may occur at tested levels (B=700g, C=60 g/L)
- Or beyond tested range (B>700g, C>60 g/L)
- Or at intermediate levels (B approximately 500g, C approximately 50 g/L)

**Conclusion:** Curvature is a significant concern. The current design cannot detect or model quadratic effects, creating uncertainty about true optimal conditions. A follow-up response surface study is strongly recommended to characterize curvature and identify actual optima.

## 11 Question 9: Follow-up Experimental Design

**Question:** Propose a follow-up experimental plan to further refine operating conditions, justifying your choice of design.

### 11.1 Answer

Based on the factorial screening results, a **Central Composite Design (CCD)** focused on factors B (inoculant size) and C (substrate concentration) is proposed for response surface optimization.

### 11.2 Design Generation

```
cat("=== CENTRAL COMPOSITE DESIGN ===\n\n")
```

```
## === CENTRAL COMPOSITE DESIGN ===
```

```
ccd.pick(k = 2)
```

```
##      n.c n0.c blks.c n.s n0.s bbr.c wbr.s bbr.s  N alpha.rot alpha.orth
## 1      4      1      1  4      1      1      1      1 10  1.414214  1.414214
## 2      4      2      1  4      2      1      1      1 12  1.414214  1.414214
## 3      4      3      1  4      3      1      1      1 14  1.414214  1.414214
## 4      4      4      1  4      4      1      1      1 16  1.414214  1.414214
## 5      4      5      1  4      5      1      1      1 18  1.414214  1.414214
## 6      4      6      1  4      6      1      1      1 20  1.414214  1.414214
## 7      4      7      1  4      7      1      1      1 22  1.414214  1.414214
## 8      4      8      1  4      8      1      1      1 24  1.414214  1.414214
## 9      4      9      1  4      9      1      1      1 26  1.414214  1.414214
## 10     4     10      1  4     10      1      1      1 28  1.414214  1.414214
```

```
cat("\n=== SELECTED DESIGN: ROTATABLE CCD ===\n\n")
```

```
##
```

```
## === SELECTED DESIGN: ROTATABLE CCD ===
```

```
ccd_design <- ccd(basis = 2,
                  n0 = c(4, 2),
                  alpha = "rotatable",
                  randomize = FALSE,
                  inscribed = FALSE)
```

```
print(ccd_design)
```

```
##      run.order std.order  x1.as.is  x2.as.is Block
## 1             1          1 -1.000000 -1.000000     1
## 2             2          2  1.000000 -1.000000     1
## 3             3          3 -1.000000  1.000000     1
## 4             4          4  1.000000  1.000000     1
```

```
## 5      5      5  0.000000  0.000000      1
## 6      6      6  0.000000  0.000000      1
## 7      7      7  0.000000  0.000000      1
## 8      8      8  0.000000  0.000000      1
## 9      1      1 -1.414214  0.000000      2
## 10     2      2  1.414214  0.000000      2
## 11     3      3  0.000000 -1.414214      2
## 12     4      4  0.000000  1.414214      2
## 13     5      5  0.000000  0.000000      2
## 14     6      6  0.000000  0.000000      2
##
## Data are stored in coded form using these coding formulas ...
## x1 ~ x1.as.is
## x2 ~ x2.as.is
```

```
cat("\n=== DESIGN COMPONENTS ===\n")
```

```
##
## === DESIGN COMPONENTS ===
```

```
cat("Total runs: 14\n")
```

```
## Total runs: 14
```

```
cat("Factorial points: 4 (corners)\n")
```

```
## Factorial points: 4 (corners)
```

```
cat("Axial points: 4 (star points)\n")
```

```
## Axial points: 4 (star points)
```

```
cat("Center points: 6 (replicates)\n")
```

```
## Center points: 6 (replicates)
```

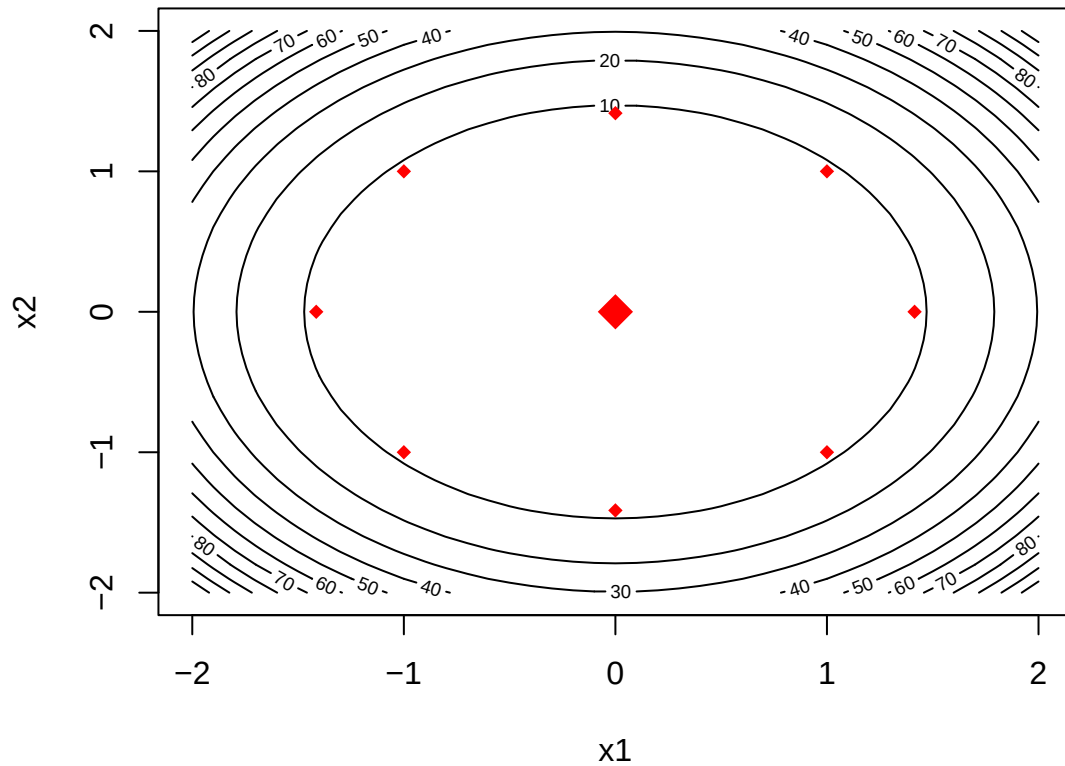
```
cat("Alpha: 1.414 (rotatable)\n")
```

```
## Alpha: 1.414 (rotatable)
```

### 11.3 Variance Function

```
varfcn(ccd_design, ~ S0(x1, x2), contour = TRUE,
       main = "Prediction Variance Function")
```

## Prediction Variance Function



The variance function shows uniform prediction precision at all points equidistant from center (rotatability property).

### 11.4 Model to be Fitted

The CCD enables fitting a second-order response surface model:

$$y = \beta_0 + \beta_1 B + \beta_2 C + \beta_{11} B^2 + \beta_{22} C^2 + \beta_{12} BC + \epsilon$$

**Model components:** - Linear effects:  $\beta_1 B, \beta_2 C$  (main effects) - Quadratic effects:  $\beta_{11} B^2, \beta_{22} C^2$  (curvature)  
- Interaction:  $\beta_{12} BC$  (already significant in factorial) - Intercept:  $\beta_0$  (estimated with center point replication)

### 11.5 Factor Ranges

```
cat("=== FACTOR CODING ===\n\n")
```

```
## === FACTOR CODING ===
```

```

cat("Factor B (Inoculant Size):\n")

## Factor B (Inoculant Size):

cat("  -1 (low):          300 g\n")

##   -1 (low):          300 g

cat("    0 (center):      500 g\n")

##    0 (center):      500 g

cat("   +1 (high):        700 g\n")

##   +1 (high):        700 g

cat("  -1.414 (axial):    ", round(500 - 200*sqrt(2), 1), " g\n", sep="")

##   -1.414 (axial):    217.2 g

cat("   +1.414 (axial):    ", round(500 + 200*sqrt(2), 1), " g\n\n", sep="")

##   +1.414 (axial):    782.8 g

cat("Factor C (Substrate Concentration):\n")

## Factor C (Substrate Concentration):

cat("  -1 (low):          40 g/L\n")

##   -1 (low):          40 g/L

cat("    0 (center):      50 g/L\n")

##    0 (center):      50 g/L

cat("   +1 (high):        60 g/L\n")

##   +1 (high):        60 g/L

cat("  -1.414 (axial):    ", round(50 - 10*sqrt(2), 1), " g/L\n", sep="")

##   -1.414 (axial):    35.9 g/L

```

Following the factorial screening, factors **B (inoculant size)** and **C (substrate concentration)** were identified as the most influential variables affecting product yield, with evidence of interaction and potential curvature. Because two-level factorial designs are limited to estimating only linear effects, they are not sufficient for accurately characterizing nonlinear behavior or identifying a true optimum in bioreactor systems.

Therefore, a **Central Composite Design (CCD)** focused on factors B and C is proposed as a follow-up experimental plan. The CCD enables estimation of quadratic and interaction effects through the inclusion of axial and center points, allowing a second-order response surface model to be fitted. This approach provides an efficient and statistically robust framework for refining operating conditions and locating the optimal region for maximum product yield.